

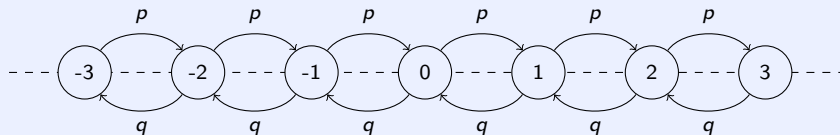
PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 28

Example

Example 1: Consider the SRW on integers ($p + q = 1$, $0 < p < 1$):



- It is easy to see that the MC is irreducible.
- Now, consider the state 0.
- Starting from 0, we can return to 0 only in even number of steps. Thus,

$$p_{00}^{(2n-1)} = 0 \quad \text{and} \quad p_{00}^{(2n)} = \binom{2n}{n} p^n q^n \quad \text{for } n \geq 1.$$

- Stirling's Formula: $n! \sim \sqrt{2\pi} n^{n+\frac{1}{2}} e^{-n}$ as $n \rightarrow \infty$.

Example

- Using the above formula,

$$p_{00}^{(2n)} = \frac{(2n)!}{(n!)^2} p^n q^n \sim \frac{1}{\sqrt{\pi}} \frac{(4pq)^n}{\sqrt{n}}.$$

- If $p = q = \frac{1}{2}$, then $4pq = 1$. Thus,

$$\sum_{n=1}^{\infty} p_{00}^{(n)} = \sum_{n=1}^{\infty} p_{00}^{(2n)} \sim \frac{1}{\sqrt{\pi}} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \infty.$$

Hence, SSRW is recurrent.

- If $p \neq q$, then $4pq < 1$. Therefore,

$$\sum_{n=1}^{\infty} p_{00}^{(n)} = \sum_{n=1}^{\infty} p_{00}^{(2n)} \sim \frac{1}{\sqrt{\pi}} \sum_{n=1}^{\infty} \frac{(4pq)^n}{\sqrt{n}} \leq \frac{1}{\sqrt{\pi}} \sum_{n=1}^{\infty} (4pq)^n < \infty.$$

Thus, asymmetric SRW is transient.

- This also shows that an infinite closed class need not be recurrent.